Sample	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	2	4
4	3	2	6
5	3	3	7

$K = 2$

1

BuildTree (5, 2)

for x_1

Split values = 1.5, 2.5

for 1.5

for x_1
split values = 1.5, 2.5

i) for 1.5
↳ Right = sample 3, 4, 5 → mean = 5.66, squared error → 4.63
↳ left = sample 1, 2 → mean = 2.5, → sq error = 0.5

ii) for 2.5
↳ Right = sample 4, 5 → mean = 6.5, → sq error = 0.5
↳ left = sample 1, 2, 3 → mean = 3, → sq error = 2

split values	sq. error
1.5	5.163
2.5	2.5

for x_2
split value = 1.5, 2.5

i) for 1.5
↳ Right ⇒ sample 2, 3, 4, 5
↓
mean = 5
sq error ⇒ 10

for left → sample 1 → mean = 2
sq error = 0

ii) for 2.5
↳ Right → sample 5 → mean = 7, sq error = 0
↳ left → sample 1, 2, 3, 4 → mean = 3.75
sq error = 8.75

feature	Split values	sum sq. er
x_1	1.5	5.13
	2.5 ✓	2.5 ✓ →
x_2	1.5	10
	2.5	8.75

Best split → 2.5

$x_1 \leq 2.5$

Sample 1, 2, 3 — 1, 2, 3 — 1, 2, 3, 4

3, 3 — 6, 7 — Sample 4, 5

Split for $(I_{j,s}, 2)$

1, 2

→ for x_1

Split values = 1.5

→ Right = sample 3 → mean = 4, sq error = 0 ✓

Left = sample 1, 2

↳ mean = 2.5 ✓

↳ sq error = 0.5

↳ $\sum_{i=1}^{n} (y - \bar{x})^2$

↳ $(2-2.5)^2 + (3-2.5)^2$

⇒

→ for x_2

split value = 1.5

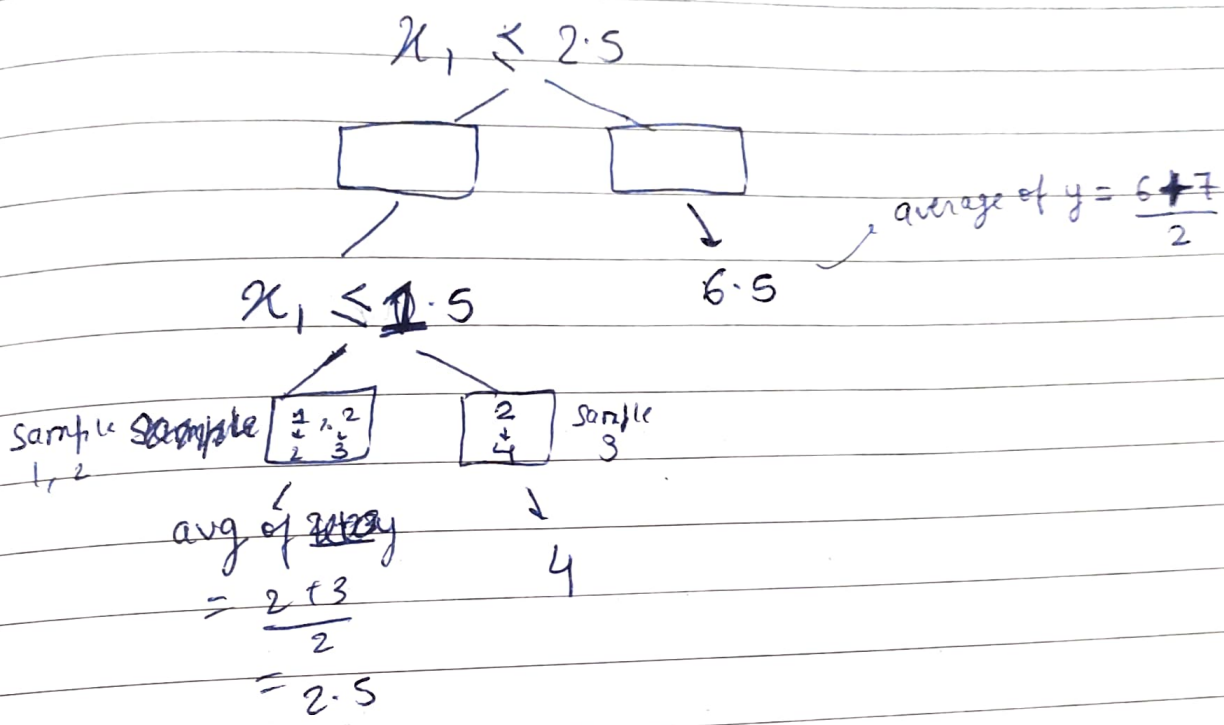
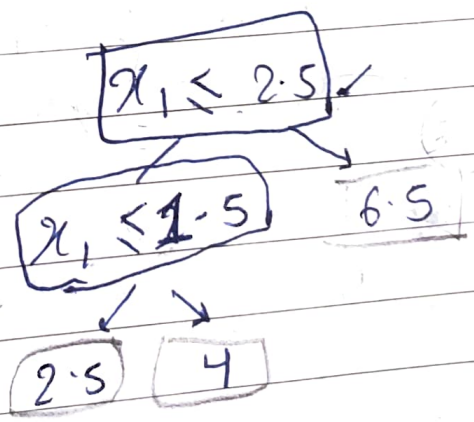
Right = sample 2, 3 → mean = 3.5

sq error = 0.5 ✓

left → sample 1 → mean = 2

sq error = 0 ✓

feature	Split value	sum sq. error
x_1	$\boxed{1.5}$	$\boxed{0.5}$ ✓
x_2	1.5	0.5



final tree :-